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中国海洋大学全日制本科课程期末考试试卷

2026 年 Spring 季学期 考试科目：Statistical Modelling and Inference

学院：海德学院

试卷类型：SAMPLE EXAM 1 命题人：Thomas Shang 课程教师：Dr. Waqar Hafeez

Examination instructions: This is a closed-book sample examination. A non-programmable calculator is permitted. Show all working. Formulae and statistical quantities required for each question are provided in the question.

Question	1	2	3	4	5	Total
Marks	20	20	20	20	20	100

Name: ____________________ Student ID: ____________________

Major/Year: ____________________ Seat Number: ____________________

Examination Room: ____________________ Instructor: ____________________

Sample Examination 1

Five questions, 20 marks each

End of cover page. Begin Question 1 on the next page.

Question 1 [20 marks total]

Bias, variance, MSE, and constructing an unbiased estimator

Let $Y_1, \ldots, Y_n$ be a random sample from a population with mean μ and variance σ^2. Consider the estimator

$$T = \frac{n-1}{n}\bar{Y}, \qquad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

For an estimator $\hat{\theta}$ of θ:

$$\mathrm{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta, \qquad \mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

Also $E(\bar{Y}) = \mu$ and $\mathrm{Var}(\bar{Y}) = \sigma^2/n$.

(a) Find the bias of T as an estimator of μ. [5 marks]

(b) Find $\mathrm{Var}(T)$. [5 marks]

(c) Find $\mathrm{MSE}(T)$. [5 marks]

(d) Define $T^* = aT$. Find a such that T^* is unbiased for μ. [5 marks]

Question 2 [20 marks total]

Confidence interval and hypothesis test for a population mean

A supermarket manager records the shopping times of $n = 64$ randomly selected customers. The sample mean is $\bar{y} = 33$ minutes. The population standard deviation is known to be $\sigma = 16$ minutes.

For known σ,

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \qquad Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}.$$

Use $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, and $\Phi(1.50) = 0.9332$.

(a) Construct a 95% confidence interval for μ. [6 marks]

(b) Test $H_0 : \mu = 30$ against $H_a : \mu > 30$ at $\alpha = 0.05$. State the test statistic and conclusion. [7 marks]

(c) Compute the one-sided p-value using the provided normal table value. [4 marks]

(d) Explain the practical conclusion in one sentence. [3 marks]

Question 3 [20 marks total]

Multiple linear regression and least squares

Five houses are observed. Price is measured in thousands of AUD, age in decades, and area in thousand square metres.

i	1	2	3	4	5
y_i	45	52	61	68	74
x_{1i} (age)	1	2	3	4	5
x_{2i} (area)	2	1	3	4	5

Fit the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$.

The least-squares normal equations are

$$(X^\top X)\hat{\beta} = X^\top y.$$

For this data set,

$$X^\top X = \begin{pmatrix} 5 & 15 & 15 \\ 15 & 55 & 56 \\ 15 & 56 & 55 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 300 \\ 978 \\ 957 \end{pmatrix}.$$

(a) Write down the design matrix X and vector y. [4 marks]

(b) Solve the normal equations for $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. [10 marks]

(c) Give the fitted regression equation. [3 marks]

(d) Predict the price for a house with age $x_1 = 3$ and area $x_2 = 4$. [3 marks]

Question 4 [20 marks total]

Parameter estimation in simple linear regression

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where X_i are fixed, $E(Y_i) = \beta_0 + \beta_1 X_i$, $\mathrm{Var}(Y_i) = \sigma^2$, and the Y_i are independent. Define

$$S_{xx} = \sum_{i=1}^{n} (X_i - \bar{X})^2, \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Use

$$\hat{\beta}_1 = \sum_{i=1}^{n} a_i Y_i, \qquad a_i = \frac{X_i - \bar{X}}{S_{xx}}, \qquad \sum a_i = 0, \qquad \sum a_i X_i = 1.$$

(a) Prove that $E(\hat{\beta}_1) = \beta_1$. [6 marks]

(b) Prove that $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$. [5 marks]

(c) Prove that $E(\hat{\beta}_0) = \beta_0$. [5 marks]

(d) State the covariance $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1)$ and explain its sign when $\bar{X} > 0$. [4 marks]

Question 5 [20 marks total]

Binomial MLE and Fisher information

Let $Y \sim \text{Binomial}(n, p)$. In one study, $n = 50$ trials give $y = 18$ successes.

The probability mass function is

$$f(y; p) = \binom{n}{y} p^y (1-p)^{n-y}.$$

For a scalar parameter p,

$$I(p) = -E\left[\frac{\partial^2 \ell(p)}{\partial p^2}\right].$$

(a) Write the likelihood and log-likelihood for p. [4 marks]

(b) Derive the score equation and find the MLE $\hat{p}$. [6 marks]

(c) Calculate the numerical MLE for the data. [3 marks]

(d) Derive the Fisher information $I(p)$ for one binomial observation with fixed n. [5 marks]

(e) Estimate the approximate standard error of $\hat{p}$ using $I(\hat{p})^{-1/2}$. [2 marks]

End of examination questions.

Check that you have attempted all five questions.